Introduction to Agile Project Management

Digital T-Level – Software Testing

Today

- Agile vs Waterfall
- Introduction to Agile approach to project management
  - Agile Manifesto
  - 12 Principles

Traditional Approach

- Start by defining exactly what the project was expected to produce.
- This was known as the requirements or specification.
  - Agreed and signed off by project team and customer
- The team then went off and created a product or system they believed to have met the requirements.

Agile Approach

- Projects start knowing that requirements will change/evolve as the project progresses.

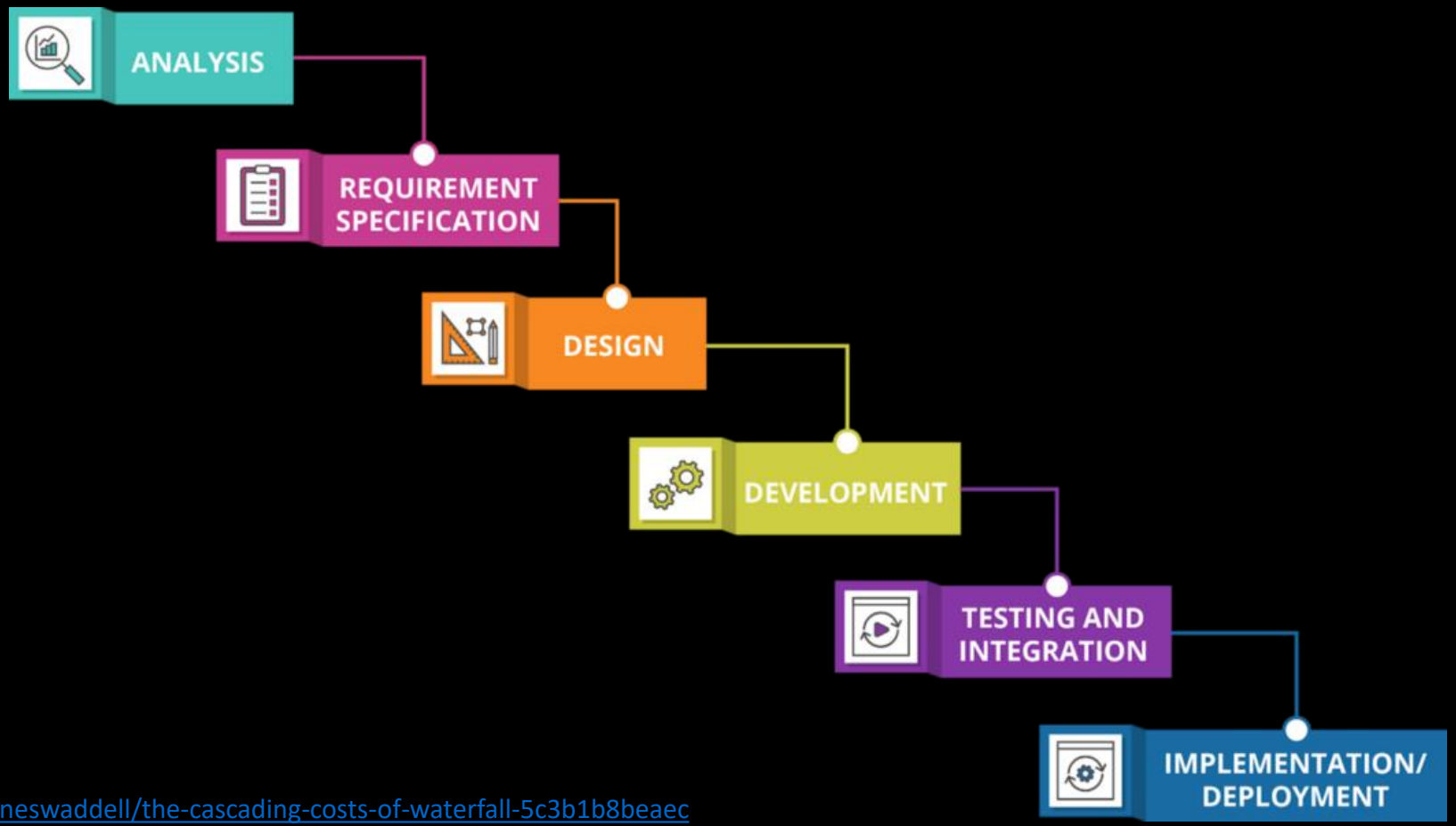
- The agreements between the project team and client are:
  - The resources that will be used
  - The time required to deliver as many as possible of the prioritised features the customer wants.

Traditional Project Methodology

- Started in the construction and manufacturing industries, where…
- …the high costs involved with making late changes meant…
- …requirements were frozen as early as possible.

- It is based on a sequential approach…
- …each stage of the project “flowed” on to the next…
- …building on the work of the preceding stage.
- This is why it led to being described as the Waterfall model…

Waterfall Model



https://medium.com/@joneswaddell/the-cascading-costs-of-waterfall-5c3b1b8beaec

Fixed Resources

- This methodology rarely suits a software design project – why?
  - Requirements typically change
  - The client wants different features
- Several attempts were made to deal with the problem
  - “Upstream propagation”
  - Changes made later in the project are fed ‘upstream’ in to the earlier stages of the project…
  - …user requirements re-written to match the features added.
- This approach had some success, but it was thought a more radical approach was needed for software projects…

Software Projects

- By the late 1990's several methodologies were in use:
  - Rapid Application Development (RAD)
  - Spiral Development
  - Iterative and Incremental Development
  - eXtreme Programming (XP)
  - Feature Driven Development (FDD)

- In all of these approaches the:
  - Requirements emerge throughout the project
  - Requirements delivered in a series of releases and…
  - …gradually build up to complete functionality.

Agile is born!

https://www.agilealliance.org/agile101/

In the late 1990's, several methodologies began to gain increasing public attention, each having a different combination of old and new ideas.

These methodologies emphasized close collaboration between the development team and business stakeholders; frequent delivery of business value, tight, self-organizing teams; and smart ways to craft, confirm, and deliver code.

The term "Agile" was applied to this collection of methodologies in early 2001 when 17 software development practitioners gathered in Snowbird, Utah to discuss their shared ideas and various approaches to software development.

This joint collection of values and principles was expressed in the Manifesto for Agile Software Development and the corresponding twelve principles.

The Agile Manifesto

Individuals and interactions	over	Processes and Tools
Working Product	over	Comprehensive Documentation
Customer Collaboration	over	Contract Negotiation
Responding to change	over	Following a plan

That is, while there is value in the items on the right, we value the items on the left more.

www.agilemanifesto.org

http://www.agilemanifesto.org

12 AGILE PRINCIPLES

01 Our highest priority is to satisfy the customer through early and continuous delivery of valuable software.	02 Welcome changing requirements, even late in development. Agile processes harness change for the customer's competitive advantage.	03 Deliver working software frequently, from a couple of weeks to a couple of months, with a preference to the shorter timescale.
04 Business people and developers must work together daily throughout the project.	05 Build projects around motivated individuals. Give them the environment and support they need, and trust them to get the job done.	06 Agile processes promote sustainable development. The sponsors, developers, and users should be able to maintain a constant pace indefinitely.
07 Working software is the primary measure of progress.	08 The most efficient and effective method of conveying information to and within a development team is face-to-face conversation.	09 Continuous attention to technical excellence and good design enhances agility.
10 Simplicity – the art of maximizing the amount of work not done – is essential.	11 The best architectures, requirements, and designs emerge from self-organizing teams.	12 At regular intervals, the team reflects on how to become more effective, then tunes and adjusts its behavior accordingly.

http://nicocasel.net/blog/wp-content/uploads/2017/06/12-agile-principles.png

Ideal Projects for Agile Methods

- Projects with unclear requirements
- Projects with a short timeline (ideally less than one year)
- Customers who can make use of iterative product delivery, with gradually enhanced features.
- Projects that can be accomplished in small teams
  - Remember agile projects value face-to-face interaction

Unsuitable Projects for Agile Methods

- Projects with clearly defined requirements, with…
- …a low likelihood of change over the course of the project.
- A project with a clear or mandated technical solution.

Agile Frameworks

- Agile is a broad term for a range of project management methods:
  - Agile Project Management
  - The Dynamic Systems Development Method (DSDM)
  - Scrum
  - eXtreme programming
  - Lean Development*
  - Feature Driven Development
  - Kanban

- Methods and ideas from other frameworks can be combined.

Assignment 1 Pointer

- We will be following the Scrum framework with regards to agile for unit 13.
- We will also be looking at the lean process of Kanban, which is regularly incorporated alongside scrum

- For assignment 1, you should be able to discuss principles of scrum and Kanban…
- …and discuss of the 10 software testing methods/phases we looked at previously fit in with them (and waterfall).